What a Cheap Judge Sees Depends on How You Ask: Perturbation-Grounded Evidence Verification from Live LLM Debates

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Abstract

TribunaLLM stages live debates between language models and checks every finished sentence against a reference library with Jev, TypeSafe's decision model, striking contradicted sentences while the next one is still arriving. Its first tournament failed as a measurement: in 180 blind debates between six frontier models the judge struck 20 of 3,357 claims, and when I altered one number in 78 sentences it had called supported, it caught none. I use the failure as an instrument. Every supported sentence with a number becomes a contrastive pair, the original and a copy with the number changed, labelled by construction: DebateCheck, 1,164 pairs split by motion, later extended with 1,657 pairs of four further perturbation types. Holding the model, the evidence and the labels fixed and varying only the question, the debate's own relation question detects 5.8% of altered sentences at its threshold, while the same criterion posed in Autorubric's binary framing [Rao and Callison-Burch, 2026b] reaches 94.7% accuracy on held-out motions. A pre-registered second study then asks whether judges err in the same places because of the task or because of the kind of judge, with six judges of three kinds on the same items and on two rubric panels with human labels. Two LLMs of different families are worth 1.16 independent votes, a decision model and an LLM more, two decision models of different makers most; part of the shared error is shared evidence; and the framing that helps Jev hurts the second decision model, so a criterion has to be tuned to the judge that answers it. Judged again with the certified question, the same 180 debates yield the leaderboard the arena promised, with 90 confident strikes instead of 20. Two of five pre-registered predictions failed and are reported.

1 Introduction

A debate is a hostile environment for a fact. Each side is rewarded for sounding right, the other side will not check the figures, and the audience cannot. TribunaLLM was built on the idea that a judge that can keep up with streaming text could change that: two language models debate a motion, every finished sentence goes to Jev, a typed decision model that answers yes/no, choice and score questions in about 200 ms for $0.042 per million input tokens, and a sentence the reference library contradicts is struck through on screen while the next one is still being written. Run headless across many models, the same engine was meant to produce a leaderboard of which model gets retracted least.

The leaderboard does not work; this paper explains why, and what the failure can be used for. Section 4 reports the pilot: 180 blind debates, 3,357 factual claims, 20 retractions, 7 of them right on review. A perturbation test settled whether the models were accurate or the judge was blind. I took 78 sentences Jev had called supported, changed one number in each (a year by a few years, a figure by a factor of two), and asked again with the same evidence: not one altered sentence

*Code, data and every intermediate file: `https://github.com/Duss02/TribunaLLM`.

was called contradicted, and 38% stayed "supported". The retrieval was not at fault (the source passage sits in the judge's top 20 for 96% of model-written claims), and Jev was not at fault as a model, because the same Jev with a different question catches the same alterations.

That observation motivates the rest of the paper. Rao and Callison-Burch [2026c] compare Jev with three LLM judges on nine rubric panels, find them wrong in the same places, trace the shared shortfall to conventions the criterion texts do not carry, and state in their limitations that their account is observational: "No experiment supplied exemplars or rewrote level descriptions." They propose rerunning their panels with exemplars and with level descriptions that name observable features. I ran an experiment of that kind, on a task where the labels are free: everything is held fixed except what the judge is told. Section 5 builds DebateCheck: every sentence a debater said that the judge called supported and that contains a number becomes a pair, the original and a copy with the number altered to one the evidence does not contain. The altered copy is contradicted by the evidence by construction; the original is the debate's own supported sentence, re-judged for consistency. 1,164 pairs, from six motions, six debaters and both sides, carry the exact twenty passages the judge read in the debate, so the judge can be re-asked offline for a few cents. Section 6 then holds the model, the evidence and the labels fixed and varies the question.

Contributions.

1. TribunaLLM, a live arena in which a decision model retracts sentences during streaming, and the negative result of its first tournament, reported in full (§4).
2. DebateCheck, a perturbation-grounded benchmark of evidence verification mined from adversarial debate, labelled by construction, split by motion, released in Autorubric's dataset format (§5).
3. The criterion-text experiment: seven ways of asking Jev the same question on identical items, from the debate's relation question (detection 5.8%) to Autorubric's framed binary criterion (accuracy 94.7%), with a decomposition into wording, evidence selection, threshold and framing (§6.1–§6.3).
4. A held-out criterion-optimisation loop for a decision model, the counterpart of Autorubric's held-out rubric improvement, which the library offers only for LLM judges (§6.2).
5. Correlated-error and cascade measurements between Jev and two reasoning-model judges on construction-labelled adversarial items (§6.4).
6. Study 2, a pre-registered test of whether judges' errors overlap because of the task or because of the kind of judge: six judges of three kinds (two decision models of different makers, three LLMs, an NLI classifier), four new perturbation types, two subjective panels with human labels, effective votes and coverage at guaranteed risk (§7).

2 Related work

Rubric-based judging and decision models. Autorubric [Rao and Callison-Burch, 2026b] makes the design choices of rubric-based LLM evaluation explicit: atomic criteria, mixed criterion types, ensembles, few-shot calibration, abstention, bias mitigations and agreement diagnostics, with decision models as judges and confidence cascades that send uncertain criteria to an LLM. I use the library as released (version 1.6.1) to validate the judges and to replicate the framing it gives a binary criterion. Rao and Callison-Burch [2026c] find Jev within noise of three LLM judges on 19 of 27 comparisons at 29 to 325 times lower cost, and find that on Jev's most confident errors the LLM judges repeat the wrong answer 96% of the time; they attribute the shared shortfall to conventions the criterion text does not state and recommend giving Jev the verdict definitions, supplying exemplars, and correcting each criterion's offset where labelled pairs exist. Rao and Callison-Burch [2026a] show that a reported agreement number depends on the judgment scale, the abstention rule and the aggregation level, and give a reporting checklist that I follow in §6.4. Li et al. [2026] measure Jev's confidence as a routing signal against sixteen judges

with blinded human adjudication: a cascade with a threshold set in advance matches GPT-6 at 41% of its fee, and the confidence tracks the boundary between verdicts that can be read from the text and verdicts that must be derived. Xu et al. [2026] show that short natural context can flip a decision model to a fixed wrong option, and Liu [2026] that a single added colon in a candidate raises Jev's false-acceptance rate from 1% to 26% under some presentations of the same request and not under others. Duan et al. [2026] benchmark decision models beyond Jev on 11,257 instances, find them competitive where a decision can be resolved from the evidence given, weaker on specialist knowledge, and prone to overstating the probability of the most likely outcome, and release open-weight decision models distilled from a reasoning LLM. On correlated errors among LLM judges, Kohli [2026] find that nine frontier judges from seven families carry about two independent votes' worth of information by the Kish effective sample size; Hossain et al. [2026] report an average pairwise error correlation of 0.21 across ten judges, equivalent to 3.5 independent judges, with the dependence strongest among the high-accuracy frontier judges, including ones from different providers; Yang et al. [2026] ask that reports of LLM-as-judge results include error-dependence estimates. These studies measure dependence among LLMs on fixed items; none varies the question or the evidence, and none puts a decision model of a second maker in the panel. My setting differs from all of these in two ways: the items are adversarial by origin, because a debater produced them under pressure to sound right, and the labels cost nothing, because I made the wrong ones myself.

Debate. Irving et al. [2018] proposed debate as a way for a weak judge to supervise strong models; Khan et al. [2024] and Kenton et al. [2024] test it with LLM debaters and judges, and Elasky et al. [2026] find that debate helps weak judges reward stronger models. Those judges decide who won. Mine decides, sentence by sentence, whether the evidence says what the debater said, which is a narrower question with a checkable answer. Public debate leaderboards [Mazur, 2026] score overall quality with LLM judges and do not verify claims.

Fact verification and contrast sets. Claim verification against evidence has a long benchmark history [Thorne et al., 2018]; Atanasova et al. [2022] build contrastive claim pairs whose labels depend on which version of the evidence is in play, the closest construction to mine. Tang et al. [2024] and Wang et al. [2023] check LLM output against grounding documents sentence by sentence offline. My pairs follow the logic of contrast sets [Gardner et al., 2020] and behavioural tests [Ribeiro et al., 2020]: a minimal edit with a known effect on the label, applied to the judge rather than to the model that wrote the text.

3 TribunaLLM: a live arena with a non-generative judge

Two debaters alternate for three rounds of at most 110 words. In *blind* mode, the one the tournament uses, they argue from what they know and see no evidence. The evidence belongs to the judge: a curated pack of 19 to 22 verified passages per motion and a reference library of 14 to 18 whole Wikipedia articles cut into passages of at most 110 words (700 to 1,700 passages, 50,000 to 120,000 words per motion). Each finished sentence is sent to Jev with the 20 library and pack passages that best match it under BM25, shuffled from a recorded seed because Jev's documentation notes a slight first-option bias. The judge never sees a model name.

The sentence request carries five typed questions over the state {motion, side, sentence, evidence}: whether the sentence is a checkable claim (`is_claim`, yes/no), whether it is supported (`supported`, yes/no), how the evidence relates to it (`relation`, a choice among *supports*, *contradicts*, *says_nothing*), which passage supports it best, and whether it is on the motion. A sentence is *contradicted* when p(contradicts) ≥ 0.7, *supported* when p(supported) ≥ 0.7, an *opinion* when p(is_claim) < 0.5, and otherwise *unverified*, which costs nothing in blind mode because silence in a library the debater never saw says nothing about the claim. Contradicted sentences are struck

Judged again (debate's question, 20 passages)	Unchanged	One number altered
supported	75 (96%)	30 (38%)
contradicted	0	0
unverified	3 (4%)	48 (62%)

Table 1: The first perturbation test, 78 sentences from the first 60 debates. The judge's own question, with the evidence it read in the debate, does not react to a wrong number.

and cost two points, supported ones earn one. A turn-level request scores rebuttal and relevance, and a final request asks which side made the stronger and more factually sound case; the three parts sum to a score and the higher score wins. Ratings are Bradley–Terry [Bradley and Terry, 1952] on wins and draws with bootstrap intervals; every pairing is played on both sides of every motion so a side advantage cancels. The thresholds were set before any data was collected and have not been tuned.

Every result line stores the full transcript: each sentence with its verdict, the three probabilities, the best passage and the ids of the twenty passages the judge read, plus the route and model id of each debater and hashes of the prompt, question and rule sources and of the pack. Everything below is computed from those files.

4 The pilot and its negative result

Six models the available keys could reach (Claude Opus 5.5 and Sonnet 5.5, GPT-6.1 Sol and GPT-6 Luna, DeepSeek V4 Pro and V4 Flash) played every pairing on both sides of six motions (nuclear power, sugar taxes, congestion pricing, a four-day week, rent control, crewed spaceflight): 180 debates, none failed, judged by `jev-1.13.0`. The debaters cost $3.10 (the two Claude models), $0.61 (the two OpenAI models) and $0.46 (the two DeepSeek models) at list prices; the judge, 8,419 calls, cost $1.10.

Retractions. Of 3,357 factual claims the judge struck 20 (0.6%). Thirteen were from one model, DeepSeek V4 Pro, and nine of those were rebuttals that deny an opponent's point which the library supports ("your claim that low-income workers get discounts is not true", when New York gives them a 50% discount). On review[1] 7 of the 20 are real contradictions, 5 are unclear and 8 are false alarms. The ranking that results (Opus, Sol and Sonnet above; Flash, Pro and Luna below, with the three intervals in each tier overlapping) is driven by the final verdict and the argument points, not by retractions, of which there are almost none.

The perturbation test. To tell a careful debater from a blind judge, I took 78 sentences the judge had called supported, changed one number in each, and asked the same five questions with the same twenty passages. Years moved by three to seven, percentages by a factor, other figures roughly doubled; the replacement was rejected if it occurred anywhere in the evidence. The unchanged sentences were judged again too, as a consistency check. Table 1 gives the result: no altered sentence was called contradicted, and 30 of 78 stayed supported, among them "in January 5,945 Mexico introduced a tax" at p(supported) = 0.93 and "Vogtle cost above $74 billion" at 0.75. The mean p(contradicts) on altered sentences was 0.15 for years, 0.30 for percentages and 0.11 for other figures, against a threshold of 0.7; no threshold rescues this, since lowering it to 0.5 finds 12 contradictions in the whole run.

Ruling out retrieval. For 300 library passages (30 per motion) a cheap model wrote one claim in its own words; BM25 over pack plus library ranked the source passage in the top 5 for 92.0% of

[1]The sample reviews in this paper were produced by a language model reading each sentence against the passages the judge read; they are not human annotations, and the review sheets are released so that a person can check them.

them, in the top 20 for 96.3%. The verbatim first sentence of a passage is found at rank 1 every time.

Other checks. Two reasoning models asked the same questions over the same evidence agree with JEV at $\kappa = 0.48$ (GPT-6.1 Sol, 103 sentences) and 0.46 (DeepSeek V4 Pro, 123), confirm one and two of its three contradictions, and find eight and eleven of their own, several genuine (5.5% where the study says 6%; 50 g CO_2/kWh where the passage says 85). The final verdict carries a position effect: listing a side first lowers its probability by 0.059 on average (standard error 0.004) over 60 debates and by 0.056 over 120 more, the opposite direction from the first-option bias the documentation describes; asking twice in the same order drifts by 0.018. The schedule plays both sides, so the effect cancels in the ranking but not in a single debate.

5 DebateCheck

Construction. From the 180 transcripts I take every sentence the judge called supported (1,920) and keep those with at least one number outside a citation (1,225). For each I build the altered copy as in the pilot test, with a fixed seed, rejecting replacements present in the evidence; 1,164 survive. A pair carries the motion, the debater and side, the ids of the twenty passages the judge read in the debate (from which the evidence is rebuilt verbatim), the kind of number altered (719 years, 140 percentages, 305 other figures) and the debate's own probabilities for the original. The split is by motion: four motions train (757 pairs), two are held out (crewed spaceflight and rent control, 407 pairs), so that anything fitted on the training motions is tested on claims about subjects it never saw.

What the labels mean. The altered sentence is wrong relative to the evidence by construction: the evidence contains the original figure and not the altered one. The original is a sentence the debate's judge called supported; my review of 30 such sentences found 22 right, 1 wrong and 7 unclear, the unclear ones mostly a figure that appears in no passage accepted because the surrounding claim is supported. So the "unchanged" side is a noisy negative, and the false-alarm rates below are upper bounds on the judge's error. The altered side is a clean positive for the question I care about, whether a judge notices that the evidence gives a different figure.

What the items are. They are not synthetic claims. A debater wrote each sentence to win an argument, under instructions that a false claim would be struck and a confirmed one rewarded. The pairs inherit the register of adversarial argument: hedges, attributions to studies, numbers quoted from memory. Table 5 shows how the six debaters differ in how often their altered sentences are caught.

6 Experiments

All runs use `jev-1.13.0` through the TypeSafe API, and the LLM judges through the providers' own APIs with reasoning effort *low*. A bootstrap over pairs (1,000 resamples) gives the intervals. Asking JEV the debate's question twice about the same unchanged sentence changes p(contradicts) by 0.035 on average and flips 15 verdicts in 1164 at the 0.7 threshold, so counts at a threshold can move by a few between runs.

6.1 Seven ways of asking

I hold the model, the items and the evidence source fixed and vary only the question (Table 2, Figure 1). The *relation* forms ask the debate's choice question; *loose* is the debate's wording (*contradicts*: "a passage states or directly implies the claim is false"), *sharp* names what a

Criterion text	AUROC [95% CI]	Det. ≥ 0.7	FA ≥ 0.7	Det. / FA ≥ 0.5	Tokens/pair
relation, loose, 20 passages (the debate's)	0.850 [0.835, 0.863]	5.8%	1.3%	11.8% / 2.0%	5,473
relation, sharp, 20 passages	0.920 [0.910, 0.930]	24.1%	0.8%	40.6% / 1.4%	5,507
relation, sharp, top-3	0.880 [0.867, 0.891]	36.9%	2.5%	52.0% / 4.0%	1,549
direct yes/no, top-3	0.865 [0.852, 0.877]	58.5%	7.3%	77.0% / 19.0%	1,399
two-step yes/no, top-3	0.728 [0.710, 0.744]	10.1%	2.7%	43.4% / 13.3%	1,461
framed criterion (Autorubric), 20 passages	0.958 [0.952, 0.964]	85.3%	7.0%	93.5% / 19.2%	5,472
framed criterion (Autorubric), top-3	0.900 [0.889, 0.911]	72.9%	9.4%	81.0% / 19.1%	1,514
mean of four relation/direct wordings	0.905 [0.895, 0.915]	21.5%	0.4%	55.4% / 4.0%	–
mean of five wordings	0.938 [0.929, 0.946]	22.3%	0.5%	63.1% / 4.1%	–

Table 2: JEV on all 1,164 pairs under each way of asking. AUROC separates altered from unchanged sentences by p(contradicts); detection is the share of altered sentences at or above the threshold, false alarm (FA) the share of unchanged ones. Tokens are the judge's input per pair (both sentences).

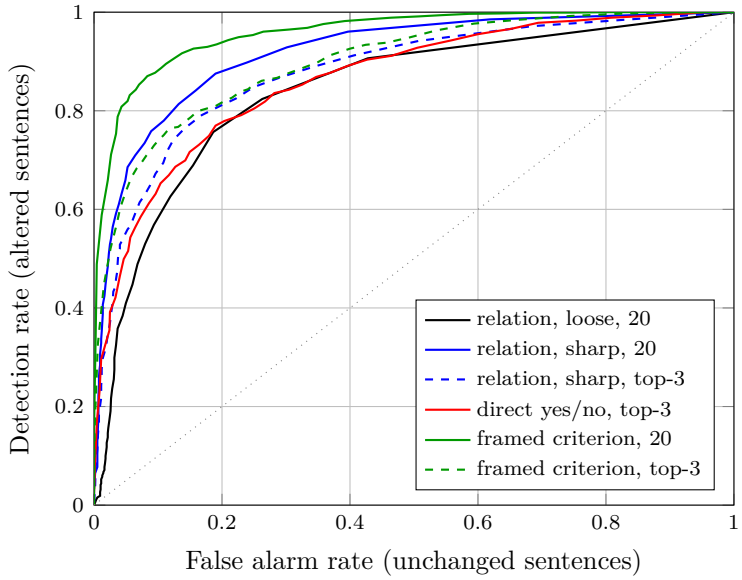


Figure 1: ROC curves of p(contradicts) for the same JEV, the same items and the same evidence source under six questions. The debate's question (black) separates the two sides of a pair almost as well as the others; what it lacks is a scale on which the altered sentences cross any sensible threshold.

contradiction looks like ("a passage gives a different number, date, name, place or outcome for the same thing `sentence` describes, or states the opposite"). "20 passages" is the evidence the judge read in the debate; "top-3" keeps the three of those twenty that best match the sentence under BM25. *Direct* asks a single yes/no question whether the evidence gives a different figure. *Two-step* asks two yes/no questions in one request, whether a passage describes the same fact and whether it gives a different value, and multiplies them. *Framed* is Autorubric's binary framing for a decision model, reproduced from the library's source: the state carries the task as `input` and the sentence as `submission`; the yes/no question reads "Determine whether this criterion is satisfied by the `submission`. Criterion: A passage in the evidence gives a different number, date, name, place or outcome for the same thing the sentence describes, or states the opposite."; and its outcomes carry the library's definitions for an error-describing criterion ("The submission advocates, states, or recommends the problematic thing" / "The submission does NOT make this error, OR mentions it only to warn against it").

Three things stand out. First, the debate's own question is not blind: its AUROC of 0.850 means that p(contradicts) ranks an altered sentence above its unchanged twin most of the time. What it lacks is location: on the training motions the mean p(contradicts) is 0.22 on altered sentences and 0.03 on unchanged ones (Table 3), apart from each other but both far below a threshold of 0.7. This is the pattern Rao and Callison-Burch [2026c] report on graded scales,

Criterion text	$\bar{p}$ altered	$\bar{p}$ unchanged	$\tau^\star$	Det.	FA	J	Det. at 0.7	FA at 0.7
relation, loose, 20 passages (the debate's)	0.22	0.03	0.04	79.9%	36.4%	0.43	6.6%	3.7%
relation, sharp, 20 passages	0.44	0.05	0.10	87.5%	24.3%	0.63	22.6%	2.2%
relation, sharp, top-3	0.49	0.09	0.20	77.6%	17.4%	0.60	39.3%	4.4%
direct yes/no, top-3	0.69	0.32	0.52	73.5%	14.5%	0.59	58.7%	6.6%
two-step yes/no, top-3	0.44	0.28	0.34	65.6%	27.0%	0.39	9.1%	3.4%
framed criterion (Autorubric), 20 passages	0.82	0.35	0.68	91.6%	4.2%	0.87	90.2%	3.9%
framed criterion (Autorubric), top-3	0.74	0.32	0.64	79.4%	6.4%	0.73	75.7%	5.2%
mean of four relation/direct wordings	0.52	0.19	0.32	80.3%	17.0%	0.63	20.4%	1.2%
mean of five wordings	0.55	0.20	0.36	86.2%	11.3%	0.75	22.4%	1.5%

Table 3: Offset correction. $\bar{p}$ is the mean p(contradicts) on the training motions; $\tau^\star$ the threshold that maximises Youden's J (detection minus false alarm) there [Youden, 1950]; the next columns apply it to the held-out motions, next to the same rates at the fixed 0.7.

Subset	n	loose, 20	sharp, 20	sharp, top-3	direct, top-3	framed, 20	framed, top-3
train motions	757	5% / 0% / 0.89	25% / 0% / 0.94	36% / 1% / 0.88	58% / 8% / 0.86	83% / 9% / 0.94	71% / 12% / 0.88
held-out motions	407	7% / 4% / 0.78	23% / 2% / 0.89	39% / 4% / 0.87	59% / 7% / 0.88	90% / 4% / 0.98	76% / 5% / 0.94
year	719	8% / 2% / 0.84	27% / 1% / 0.92	38% / 3% / 0.90	61% / 8% / 0.89	89% / 8% / 0.97	81% / 9% / 0.93
percent	140	1% / 1% / 0.90	25% / 0% / 0.97	34% / 2% / 0.83	61% / 12% / 0.80	85% / 7% / 0.95	62% / 17% / 0.80
other	305	4% / 0% / 0.84	18% / 0% / 0.91	35% / 2% / 0.86	50% / 5% / 0.85	77% / 6% / 0.94	59% / 6% / 0.88

Table 4: Detection at 0.7 / false alarm at 0.7 / AUROC by split and by the kind of number altered.

where the judges' ordering is right and their placement is off by a level, now on a binary verdict. Second, wording and evidence selection do different jobs. The sharp wording raises AUROC on the full evidence to 0.920, the best ranking of any relation form, while narrowing the evidence to three passages raises detection at the fixed threshold to 36.9% by moving the scale up, at some cost in ranking. Third, the framed criterion changes the outcome qualitatively: with the same twenty passages, AUROC is 0.958, detection at 0.7 is 85.3% at 7.0% false alarms over all pairs, and on the held-out motions 90.2% at 3.9% (AUROC 0.980). For this form more evidence is better (0.958 with twenty passages against 0.900 with three), the reverse of the relation form, where narrowing the evidence was the only way to push the scale up. Nothing in the framed question that the relation question lacks is information about the task; what it adds is the verdict definitions and a task sentence that tells the model what the two fields are for, which is the first row of the recommendations in Rao and Callison-Burch [2026c].

Correcting the offset. Where labelled pairs exist, Rao and Callison-Burch [2026c] suggest correcting each criterion's offset. My labels are free, so I fit the threshold on the training motions and apply it to the held-out ones (Table 3). Under the debate's wording the fitted threshold is 0.04, which recovers 79.9% detection at 36.4% false alarms: the information was there, at a scale no one would have set by hand. The sharper wordings fit higher thresholds and keep more of their detection at fewer false alarms, so wording and offset are complements, not substitutes.

Kinds and debaters. Table 4 splits by the number altered. Years are the easiest to catch and percentages the hardest under most wordings; the held-out motions behave like the training ones. Table 5 splits by the model that wrote the sentence: the spread between debaters in how often an altered sentence is caught is wider than the spread in how often an unchanged one is flagged, which says that some models write sentences whose numbers are easier to check against a library (a date, a named study) and others write sentences whose numbers float free of any passage.

6.2 Learning the criterion text on held-out motions

Autorubric's `held_out` strategy revises a criterion's wording from its grading errors on labelled data, but only for LLM judges; decision models are excluded. I implement the loop for Jev (Table 6): starting from the sharp wording over the top-3 passages, each round judges a fixed

Debater	Pairs	Loose det.	Sharp-3 det.	Sharp-3 FA	Sharp-3 AUROC
claude-opus-5.5	250	9.2%	45.6%	1.6%	0.898
claude-sonnet-5.5	187	3.2%	32.1%	2.1%	0.879
deepseek-v4-flash	231	3.9%	28.6%	4.3%	0.811
deepseek-v4-pro	161	2.5%	31.7%	4.3%	0.839
gpt-6-luna	134	6.0%	34.3%	1.5%	0.892
gpt-6.1-sol	201	8.5%	46.3%	1.0%	0.944

Table 5: By debater: how often its altered sentences are caught (detection at 0.7) and how often its unchanged ones are flagged, under the debate's wording and the sharp top-3 wording.

Wording	Train AUROC	Train det.	Train FA	Held-out AUROC	Held-out det.	Held-out FA
Seed (sharp)	0.899	37.3%	0.8%	0.873	40.0%	3.9%
Optimised	0.905	44.3%	1.0%	0.893	49.9%	4.2%

Table 6: Held-out criterion optimisation. The training columns are on the 400-pair sample the loop saw; the held-out columns are on the 407 pairs from motions it never saw.

sample of 400 training pairs, builds an error report with the eight altered sentences the model missed most confidently and the eight unchanged ones it flagged most confidently, each with its passages, and asks a reviser model (gpt-6.1-sol) for four alternative descriptions of the *contradicts* option; the best by training AUROC is kept. Only the text of one option changes; the form, the other two options and the evidence stay fixed. After 3 rounds and 12 candidates ($0.40 of Jev time), the winner is measured once on the held-out motions.

The optimised wording generalises: AUROC on the held-out motions goes from 0.873 to 0.893, detection at 0.7 from 40.0% to 49.9%, false alarms from 3.9% to 4.2%. The winning text is: *"The 'evidence' gives a different number, date, name, place, threshold, or outcome for a fact asserted in 'sentence', or explicitly states its opposite, even if the rest of the sentence agrees with the evidence."*. The appendix lists every wording tried. The gain is real but smaller than the gain from changing the framing, which suggests that the loop should search over the form of the question as well as its words.

6.3 Framing through Autorubric

To check that the framed result is a property of the question and not of my code, I exported the held-out split as an Autorubric `RubricDataset` (one item per sentence, the motion and the evidence in the item's prompt, one negative-weight criterion that is MET when the evidence contradicts the sentence, ground truth MET for altered and UNMET for unchanged) and graded it with the library's `DecisionModelConfig`, under both of its binary framings, and with its metrics (Table 7). Nothing in the item names its label; the library sends only the prompt, the submission and the questions.

With all twenty passages, Jev reaches 93.9% accuracy under the yes/no framing and 94.7% under the three-way choice ($\kappa = 0.89$), with recall 97.3% and 94.8% on altered sentences. With only the three best-matching passages it drops to 86.0%, still above GPT-6 Luna graded by the same library on the same three passages (82.8%, $\kappa = 0.66$). Under the three abstention rules of Rao and Callison-Burch [2026a] the choice framing's accuracy moves by less than a point, because it abstains on four items of 814. The same framing reproduced in my own client (*framed* in Table 2) gives detection 93.5% at 0.5 on all pairs, so the effect is the question, not the harness. This is, as far as I can tell, the largest single lever in the study: the relation question that the arena ran for 180 debates and the framed criterion are the same model reading the same passages about the same sentence, and one of them catches a wrong number nine times in ten while the other catches it once in twenty.

Evidence	Judge (Autorubric)	Items	Acc. [95% CI]	κ [95% CI]	ϕ	Prec.	Rec.	TP/FP/FN/TN
top-3	jev-noul_framed	814	85.9% [0.83, 0.88]	0.72 [0.67, 0.77]	0.72	88%	83%	338/46/69/361
top-3	jev-choice	814	86.0% [0.84, 0.88]	0.73 [0.68, 0.77]	0.74	91%	82%	335/35/72/372
top-3	llm-openai/gpt-6-luna	814	82.8% [0.80, 0.85]	0.66 [0.60, 0.71]	0.66	83%	82%	334/67/73/340
20 passages	jev-noul_framed	814	93.9% [0.92, 0.95]	0.88 [0.84, 0.91]	0.88	91%	97%	396/39/11/368
20 passages	jev-choice	814	94.7% [0.93, 0.96]	0.89 [0.86, 0.92]	0.90	95%	95%	386/21/21/386

Table 7: The held-out split graded with the Autorubric library as released. `noul_framed` is a yes/no question with the MET/UNMET definitions; `choice` offers MET, UNMET and CANNOT_ASSESS. Intervals are the library's bootstrap over items.

Judge	Items	Cov.	p_{judge}	p_{label}	Acc.	ϕ	κ [95% CI]	Prec.	Rec.	TP/FP/FN/TN
Jev, loose, 20, $p \geq 0.7$	2328	100%	3.5%	50%	52.2%	0.12	0.04 [0.03, 0.06]	82%	6%	67/15/1097/1149
Jev, sharp, 20, $p \geq 0.7$	2328	100%	12.5%	50%	61.7%	0.35	0.23 [0.21, 0.26]	97%	24%	281/9/883/1155
Jev, sharp, top-3, $p \geq 0.7$	2328	100%	19.7%	50%	67.2%	0.43	0.34 [0.32, 0.37]	94%	37%	430/29/734/1135
Jev, sharp, top-3, $p \geq 0.5$	2328	100%	28.0%	50%	74.0%	0.53	0.48 [0.45, 0.51]	93%	52%	605/47/559/1117
Jev, direct, top-3, $p \geq 0.7$	2328	100%	32.9%	50%	75.6%	0.54	0.51 [0.48, 0.54]	89%	59%	681/85/483/1079
Jev, two-step, top-3, $p \geq 0.7$	2328	100%	6.4%	50%	53.7%	0.15	0.07 [0.05, 0.09]	79%	10%	117/32/1047/1132
Jev, framed, 20, $p \geq 0.5$	2328	100%	56.3%	50%	87.2%	0.75	0.74 [0.72, 0.77]	83%	93%	1088/223/76/941
Jev, framed, top-3, $p \geq 0.5$	2328	100%	50.0%	50%	81.0%	0.62	0.62 [0.59, 0.65]	81%	81%	943/222/221/942
Jev, mean of 4, $p \geq 0.5$	2328	100%	29.7%	50%	75.7%	0.56	0.51 [0.49, 0.54]	93%	55%	645/47/519/1117
deepseek-v4-flash (sharp, top-3)	2080	89%	46.3%	52%	83.5%	0.67	0.67 [0.64, 0.70]	88%	79%	851/113/231/885
gpt-6-luna (sharp, top-3)	2328	100%	41.0%	50%	82.0%	0.65	0.64 [0.61, 0.67]	89%	73%	849/105/315/1059

Table 8: Every judge as a binary contradiction detector on the same items. Coverage is the share of items with a usable answer. p_{judge} and p_{label} are the positive rates; the label rate is 50% by construction.

6.4 Reasoning-model judges, correlated errors and cascades

Two reasoning models of different families, GPT-6 Luna and DeepSeek V4 Flash, answered the sharp relation question over the same top-3 passages as Jev's *sharp, top-3* form, as JSON, at reasoning effort *low* and temperature 0. Table 8 reports every judge as a binary detector under the protocol of Rao and Callison-Burch [2026a]: two items per pair, the altered one positive; an LLM answer that did not parse is an abstention, excluded and counted in coverage; both positive rates beside κ and ϕ; the 2×2 counts; pairs as the resampling unit.

Correlated errors. Table 9 asks the question of Rao and Callison-Burch [2026c] on construction-labelled items. A judge that errs more also errs on easier items, where the other judge is right, so the fair comparison matches evidence and error rate. Under the framed wording over the same three passages the LLMs read, Jev errs on 19.0% of items and GPT-6 Luna on 18.0%. On the items Jev gets wrong, Luna is wrong on 60.5%, 3.35 times what independence predicts ($\kappa_{\text{err}} = 0.53$ between the two error indicators); DeepSeek V4 Flash is wrong on 67.0% of them ($\kappa_{\text{err}} = 0.63$). The two LLM judges, at the same error rates, fail on each other's items far more often: Flash is wrong on 86.3% of the items Luna is wrong on, 5.10 times independence ($\kappa_{\text{err}} = 0.82$). Under the unmatched relation wording, where Jev errs on 26.0% of items, its overlap with Luna is 50.8%, and on Jev's 305 confident errors 67.5%. So the dependence between Jev and an LLM is real, smaller than between two LLMs, and smaller than the 96% repetition rate Rao and Callison-Burch [2026c] report on rubric panels, where every judge placed essays one level too low together. Part of what the judges share here is plain difficulty: on the items DeepSeek V4 Flash gets wrong (344 items), Jev errs on 70.9% and Luna on 86.3%, and Luna is wrong on 92.6% of Jev's errors; on the items Flash gets right (1736), Jev errs on 6.9%, Luna on 3.2%, and the overlap is 8.3%. A share of the pairs is hard for every judge, typically a figure quoted from an edition the pack does not hold or a passage about a neighbouring quantity; outside it, Jev and the LLMs fail on different sentences and for different reasons, Jev's errors being misses on a compressed scale and the LLMs' being readings. Lower overlap is only worth something if a combination cashes it in, so I tried (held-out motions, 719 items with answers from every judge; rules fitted on the training motions). If one could pick the right judge for each item, Jev with

Jev wording	LLM judge	Items	LLM err.	Jev err.	P(LLM wrong | Jev wrong)	P(LLM wrong | Jev conf. wrong)	n conf.	κ_{err}
sharp, top-3	deepseek-v4-flash	2080	16.5%	25.5%	44.3%	61.2%	263	0.42
sharp, top-3	gpt-6-luna	2328	18.0%	26.0%	50.8%	67.5%	305	0.49
framed, top-3	deepseek-v4-flash	2080	16.5%	17.5%	67.0%	88.0%	117	0.63
framed, top-3	gpt-6-luna	2328	18.0%	19.0%	60.5%	78.2%	133	0.53
framed, 20	deepseek-v4-flash	2080	16.5%	11.9%	49.4%	67.6%	37	0.32
framed, 20	gpt-6-luna	2328	18.0%	12.8%	40.8%	48.9%	47	0.22

Table 9: Correlated errors. An error is a wrong binary decision on an item; Jev decides at 0.5 under the wording in the first column, and a confident Jev error has $p \leq 0.2$ on an altered sentence or $p \geq 0.8$ on an unchanged one. The independence baseline for each conditional rate is the LLM's own error rate in the third column. κ_{err} is Cohen's κ between the two judges' error indicators.

GPT-6 Luna would reach 91.2% against 89.0% for the two LLMs. No rule I fitted gets there: a weighted vote of Jev and Luna scores 88.5% where the better of the two alone scores 88.2%, the two LLMs together 87.8% against 87.5%, and a majority of all three 88.7%. The items the judges disagree on are the hard ones, and a vote cannot tell who is right. Two different questions to the same Jev are, moreover, as complementary as Jev and an LLM (90.7% if one could pick), so what the lower overlap reflects is different questions, not a different kind of judge. On these items a second judge buys little accuracy; what buys accuracy is asking the first judge well: Jev alone with twenty passages scores 94.4% on the same items. A caveat on the comparison in Table 8: the LLM judges read the three best-matching passages, the same evidence as Jev's *sharp, top-3* and *framed, top-3* forms; on those three passages framed Jev ($\kappa = 0.62$) is within noise of GPT-6 Luna (0.64) and DeepSeek V4 Flash (0.67) at a quarter and a twenty-fifth of their cost, and with the twenty passages the LLMs were not given it pulls ahead (0.74).

Confidence and cascades. Li et al. [2026] and Rao and Callison-Burch [2026b] route uncertain items from Jev to an LLM. I simulate the cascade on these items: Jev decides when $|2p - 1| \geq \tau$ and the LLM decides otherwise, with cost at list prices from the tokens each judge used (Table 10, Figure 2). The AUROC of Jev's confidence for its own correctness is 0.698 under the sharp wording and 0.834 under the framed one, in the range Li et al. [2026] report. Under the sharp wording Jev alone is at 74.0% and GPT-6 Luna at 82.0%; the cascade that stays within one point of the LLM escalates 38.6% of items at 64% of the LLM's cost (38.8% and 43% with DeepSeek V4 Flash as the fallback). Under the framed wording Jev alone (87.2%) is above both LLM judges, so the best cascade escalates nothing. Autorubric's own calibration of a cascade on the held-out split with three passages, through its `calibrate_escalation`, picks a threshold of 0.06 that escalates 1.8% of items for 86.2% accuracy, against 82.8% for the LLM alone and 85.9% for Jev alone. As in Rao and Callison-Burch [2026c], a cascade is a cost tool: it buys the LLM's accuracy at a fraction of its price and cannot add what both judges lack. One more caveat on price: a decision model that reads twenty passages ($0.000115 per item at list price) costs about as much as GPT-6 Luna reading three ($0.00014), because that model is itself priced near the floor; against DeepSeek V4 Flash with reasoning ($0.00081) the gap is sevenfold.

Ensembles of wordings. Averaging p(contradicts) over four relation and direct wordings gives AUROC 0.905, above any single relation wording, at four times the price; at a threshold of 0.5 it detects 55.4% at 4.0% false alarms. Adding the framed wording to the mean gives 0.938, still below the framed criterion asked once (0.958). A decision model's answer varies with the wording, so asking several ways is a cheap test-time ensemble, but averaging a good question with worse ones dilutes it.

7 Study 2: where judges err together

The correlated-error result of §6.4 invites a question it cannot answer: is the weaker overlap between Jev and an LLM a property of the kind of judge, or of a task whose labels are free and

Jev wording	Fallback	τ	Escalated	Accuracy	Cost/item	Share of LLM cost
sharp, top-3	deepseek-v4-flash	0.0	0.0%	74.5%	$0.00003	4%
sharp, top-3	deepseek-v4-flash	0.4	17.0%	78.2%	$0.00017	21%
sharp, top-3	deepseek-v4-flash	0.6	30.1%	81.5%	$0.00028	35%
sharp, top-3	deepseek-v4-flash	0.8	47.6%	83.0%	$0.00043	53%
sharp, top-3	deepseek-v4-flash	1.0	91.4%	83.4%	$0.00081	99%
sharp, top-3	gpt-6-luna	0.0	0.0%	74.0%	$0.00003	24%
sharp, top-3	gpt-6-luna	0.4	16.8%	77.4%	$0.00006	41%
sharp, top-3	gpt-6-luna	0.6	29.9%	80.3%	$0.00007	55%
sharp, top-3	gpt-6-luna	0.8	47.7%	82.0%	$0.00010	74%
sharp, top-3	gpt-6-luna	1.0	92.1%	82.0%	$0.00016	118%
framed, top-3	deepseek-v4-flash	0.0	0.0%	82.5%	$0.00003	4%
framed, top-3	deepseek-v4-flash	0.4	18.9%	84.1%	$0.00026	32%
framed, top-3	deepseek-v4-flash	0.6	31.9%	83.9%	$0.00039	48%
framed, top-3	deepseek-v4-flash	0.8	63.2%	83.6%	$0.00065	80%
framed, top-3	deepseek-v4-flash	1.0	100.0%	83.5%	$0.00084	104%
framed, top-3	gpt-6-luna	0.0	0.0%	81.0%	$0.00003	24%
framed, top-3	gpt-6-luna	0.4	20.6%	83.2%	$0.00006	48%
framed, top-3	gpt-6-luna	0.6	34.8%	83.2%	$0.00009	63%
framed, top-3	gpt-6-luna	0.8	66.1%	82.5%	$0.00013	95%
framed, top-3	gpt-6-luna	1.0	100.0%	82.0%	$0.00017	124%
framed, 20	deepseek-v4-flash	0.0	0.0%	88.1%	$0.00011	14%
framed, 20	deepseek-v4-flash	0.4	22.5%	89.8%	$0.00035	44%
framed, 20	deepseek-v4-flash	0.6	36.7%	89.0%	$0.00050	61%
framed, 20	deepseek-v4-flash	0.8	68.4%	86.1%	$0.00075	92%
framed, 20	deepseek-v4-flash	1.0	100.0%	83.5%	$0.00093	114%
framed, 20	gpt-6-luna	0.0	0.0%	87.2%	$0.00011	85%
framed, 20	gpt-6-luna	0.4	24.0%	89.6%	$0.00015	112%
framed, 20	gpt-6-luna	0.6	39.0%	89.3%	$0.00017	128%
framed, 20	gpt-6-luna	0.8	70.0%	85.7%	$0.00021	159%
framed, 20	gpt-6-luna	1.0	100.0%	82.0%	$0.00025	185%

Table 10: Cascades at selected confidence thresholds: items with Jev confidence below τ go to the LLM. Cost per item at list prices; the last column is the cascade's cost as a share of the LLM's.

whose evidence is fixed? Rao and Callison-Burch [2026c] measured overlap on subjective rubric panels with human labels; I measured it on construction-labelled contradiction pairs. The two differ in task, evidence, judges and question at once. Study 2 separates them. It was pre-registered before any new API call (hypotheses, items, judges, measures, stopping rule; the pre-registration and its deviations ship with the code), and every result is reported whether or not it matches the prediction.

7.1 Hypotheses

H1 *Task objectivity.* For the same pair of judges, the correlation of errors (ϕ between the two error indicators) is lower on DebateCheck than on a subjective panel with human labels. Support: lower for at least 80% of pairs, intervals disjoint for at least half.

H2 *Kind of judge.* On DebateCheck, with identical evidence, error correlation is ordered: two LLMs > a decision model with an LLM > two decision models of different makers (Jev and GPT-6 Luna through OpenAI's Decisions API). Stated before any Luna Decisions data existed.

H3 *Error type.* Jev's recall on altered entities, attributions, polarities and units is lower than on altered numbers by at least 15 points; GPT-6 Luna's falls by less than 10.

H4 *Difficulty.* Within strata defined by whether a third judge was right, Jev–LLM correlation stays below LLM–LLM correlation.

H5 *Framing replicates.* The framed criterion's AUROC gain over the relation question (for Jev,

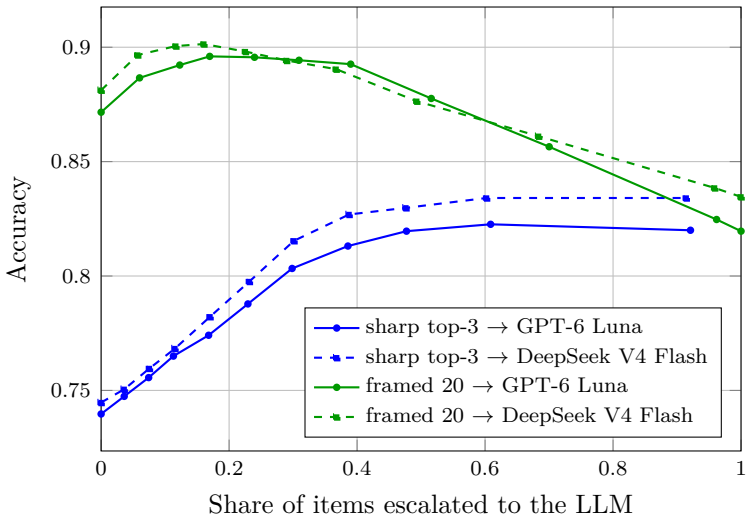


Figure 2: Cascade accuracy against the share of items escalated, sweeping the confidence threshold from 0 (Jev alone, left) to 1 (the LLM alone, right).

§6.3) replicates for GPT-6 Luna Decisions, with a gain of at least 0.05.

7.2 Items, judges and measures

Items. The 1,164 numeric pairs of §5 are unchanged. To them I add 1,657 pairs of four new types, built from the same supported sentences with the same rule (the original token occurs in the evidence the judge reads, the replacement occurs nowhere in it, one edit per sentence, seed 2026): *polarity* (a direction word flipped: rose/fell, more/fewer, higher/lower; 269 pairs), *entity* (a country or city replaced by one of the same kind absent from the evidence; 741), *attribution* (the source named for a fact replaced by another of the same kind; 305) and *unit* (percent/percentage points, million/billion, per year/per month, dollars/euros; 342). Direction words are too common for the whole-evidence rule, so for polarity the rule is applied to the passages that state the sentence's fact; this amendment was recorded before any judge ran. The split by motion is unchanged (held-out: crewed spaceflight, rent control; 407 numeric and 400 new-type pairs). Every candidate edit is checked by code and rejections are logged.

Judges. Every judge answers the same framed criterion (§6.3) over the same evidence text, in two conditions: the twenty passages the arena read and the three that best match the sentence by BM25. Six judges of three kinds: two decision models of different makers, Jev 1.13.0 and GPT-6 Luna through the Decisions API (a predicate question, probability returned, input tokens only billed); three LLM judges answering with a JSON verdict at reasoning effort *low*, GPT-6 Luna, DeepSeek V4 Pro and Claude Sonnet 5.5 (the last on a seeded sample of 400 held-out pairs, 200 numeric and 50 per new type; DeepSeek on every numeric pair and 1,262 of the 1,657 new-type pairs, where its balance ran out); and an NLI consistency classifier, Vectara's HHEM-2.1-Open, run on the CPU with the three passages as premise. Decision models and HHEM decide at 0.5, with a threshold fitted on the training motions reported beside it; an LLM answer without a parsable verdict is an abstention, excluded and counted in coverage.

Subjective panels. For H1 the same judges grade two rubric panels with human labels through the Autorubric library: CHARM-100 (the library's own example, 100 conversations, six criteria; primary) and HealthBench-100 test (100 items with physician majority labels per criterion; secondary), both obtained before any judge ran on them. Ordinal criteria are read as MET in the upper half of their scale; the nominal length criterion and not-applicable labels are not used. Jev runs under the library's `noul_framed` framing, the LLM judges through LiteLLM at thinking level

judge	n	judge positive	accuracy	recall (altered)	false alarm	AUROC
Jev, framed, 20 passages	1,614	46.2%	86.1% [84.6%, 87.6%]	82.3% [79.7%, 84.9%]	10.2% [8.1%, 12.3%]	0.916 [0.904, 0.928]
Jev, framed, top-3	1,614	42.7%	78.2% [76.4%, 80.0%]	70.9% [67.7%, 74.1%]	14.5% [12.1%, 16.9%]	0.870 [0.855, 0.885]
Luna Decisions, framed, 20 passages	1,614	57.4%	70.2% [67.8%, 72.6%]	77.6% [74.6%, 80.4%]	37.2% [33.7%, 40.5%]	0.775 [0.751, 0.798]
Luna Decisions, framed, top-3	1,614	49.9%	66.2% [64.0%, 68.5%]	66.0% [62.7%, 69.3%]	33.7% [30.4%, 36.8%]	0.726 [0.704, 0.749]
Luna chat, framed, top-3	1,614	48.3%	79.1% [77.1%, 80.9%]	77.3% [74.2%, 80.2%]	19.2% [16.7%, 21.9%]	–
DeepSeek V4 Pro, framed, top-3	1,221	50.4%	81.2% [79.0%, 83.3%]	81.3% [78.2%, 84.4%]	18.8% [15.8%, 22.3%]	–
Claude Sonnet 5.5, framed, top-3	800	45.9%	79.1% [76.6%, 81.9%]	75.0% [70.5%, 79.0%]	16.8% [13.3%, 20.8%]	–
HHEM-2.1-Open, top-3	1,614	68.0%	66.5% [64.6%, 68.2%]	84.5% [81.9%, 86.9%]	51.4% [48.1%, 54.8%]	0.730 [0.710, 0.748]

Table 11: Every judge of Study 2 as a binary contradiction detector on the held-out motions, all five perturbation types pooled (the label rate is 50% by construction). Claude Sonnet 5.5 was run on 400 of the 807 held-out pairs.

judge	number	polarity	entity	attribution	unit
Jev, framed, 20 passages	96.8%	90.0%	81.6%	66.0%	36.1%
Jev, framed, top-3	83.8%	84.0%	69.4%	59.4%	24.7%
Luna Decisions, framed, 20 passages	83.0%	76.0%	76.2%	73.6%	61.9%
Luna Decisions, framed, top-3	69.8%	86.0%	61.2%	64.2%	49.5%
Luna chat, framed, top-3	81.6%	88.0%	78.9%	70.8%	58.8%
DeepSeek V4 Pro, framed, top-3	83.4%	86.2%	87.7%	80.0%	52.1%
Claude Sonnet 5.5, framed, top-3	79.5%	92.0%	76.0%	82.0%	32.0%
HHEM-2.1-Open, top-3	94.3%	40.0%	96.6%	82.1%	50.5%

Table 12: Recall on altered sentences by perturbation type, held-out motions, decision at 0.5 for the scored judges.

low, Luna Decisions with one typed question per criterion. HHEM answers no question on these panels (no criterion compares the submission with a source), DeepSeek's API rejected the library's structured-output format on the day of the run, and Claude was stopped after CHARM-100 by the pre-registered spending rule; the pre-registration lists these deviations.

Measures. Per judge: accuracy, recall on altered sentences, false-alarm rate on unchanged ones, AUROC where a probability exists, overall and by type. Per pair of judges, on the items both answered: the 2×2 table of error indicators, ϕ, Cohen's κ, $P(B \text{ wrong} \mid A \text{ wrong})$ against $P(B \text{ wrong})$, and the effective number of independent votes $k/(1 + (k-1)\bar{\phi})$, the design-effect correction for equicorrelated units [Kish, 1965]. Coverage at guaranteed risk: thresholds chosen on the training motions as the widest acceptance region whose one-sided Clopper–Pearson 95% bound on the error is at most 5%, evaluated on the held-out motions. Intervals are percentile bootstraps resampling pairs (conversations on the panels), 1,000 resamples. Thresholds, selective rules and nothing else are fitted on the training motions; the held-out motions were not opened until every judge had finished on them.

7.3 Results on the held-out motions

Judges and error types (H3). Table 11 pools the five types; Table 12 splits recall by type. Jev under the framed criterion over twenty passages is the most accurate judge on the held-out motions (86.1%, AUROC 0.916), ahead of the three LLM judges (79.1% to 81.2%), of GPT-6 Luna Decisions (70.2%) and of HHEM (66.5%, with 51.4% false alarms); its advantage is a false-alarm rate (10.2%) about half the LLMs' and a quarter of the second decision model's. H3 is supported: from numbers to the mean of the four new types, Jev's recall over three passages falls by 24.4 points (28.4 over twenty), GPT-6 Luna Decisions' by 4.6 (11.1), Luna chat's by 7.5, DeepSeek V4 Pro's by 6.9, Claude's by 9.0 and HHEM's by 27.1. The drop is not uniform: Jev catches a flipped direction word as often as a changed number (90.0%), a changed place or country most of the time (81.6%), a changed source two times in three (66.0%) and a changed unit one time in three (36.1%), where every judge is weak and the human check of the construction labels (§7,

pair	n	both / A / B / neither wrong	ϕ	κ	$P(B \mid A)$ vs $P(B)$	lift
Jev, framed, top-3 × Luna Decisions, framed, top-3	1,614	247 / 105 / 299 / 963	0.406 [0.357, 0.458]	0.388 [0.339, 0.440]	70.2% vs 33.8%	2.07
Jev, framed, top-3 × Luna chat, framed, top-3	1,614	207 / 145 / 131 / 1131	0.491 [0.439, 0.543]	0.491 [0.439, 0.543]	58.8% vs 20.9%	2.81
Jev, framed, top-3 × DeepSeek V4 Pro, framed, top-3	1,221	142 / 89 / 87 / 903	0.529 [0.461, 0.591]	0.529 [0.461, 0.591]	61.5% vs 18.8%	3.28
Jev, framed, top-3 × Claude Sonnet 5.5, framed, top-3	800	117 / 62 / 50 / 571	0.588 [0.512, 0.656]	0.587 [0.511, 0.656]	65.4% vs 20.9%	3.13
Jev, framed, top-3 × HHEM-2.1-Open, top-3	1,614	124 / 228 / 416 / 846	0.020 [-0.034, 0.071]	0.019 [-0.032, 0.069]	35.2% vs 33.5%	1.05
Luna Decisions, framed, top-3 × Luna chat, framed, top-3	1,614	232 / 314 / 106 / 962	0.379 [0.325, 0.432]	0.359 [0.307, 0.412]	42.5% vs 20.9%	2.03
Luna Decisions, framed, top-3 × DeepSeek V4 Pro, framed, top-3	1,221	158 / 238 / 71 / 754	0.375 [0.313, 0.440]	0.351 [0.290, 0.416]	39.9% vs 18.8%	2.13
Luna Decisions, framed, top-3 × Claude Sonnet 5.5, framed, top-3	800	114 / 155 / 53 / 478	0.377 [0.300, 0.452]	0.357 [0.284, 0.433]	42.4% vs 20.9%	2.03
Luna Decisions, framed, top-3 × HHEM-2.1-Open, top-3	1,614	159 / 387 / 381 / 687	-0.066 [-0.114, -0.020]	-0.066 [-0.114, -0.020]	29.1% vs 33.5%	0.87
Luna chat, framed, top-3 × DeepSeek V4 Pro, framed, top-3	1,221	180 / 53 / 49 / 939	0.728 [0.677, 0.776]	0.728 [0.677, 0.775]	77.3% vs 18.8%	4.12
Luna chat, framed, top-3 × Claude Sonnet 5.5, framed, top-3	800	124 / 45 / 43 / 588	0.668 [0.602, 0.731]	0.668 [0.601, 0.731]	73.4% vs 20.9%	3.51
Luna chat, framed, top-3 × HHEM-2.1-Open, top-3	1,614	119 / 219 / 421 / 855	0.019 [-0.033, 0.069]	0.018 [-0.031, 0.065]	35.2% vs 33.5%	1.05
DeepSeek V4 Pro, framed, top-3 × Claude Sonnet 5.5, framed, top-3	612	93 / 25 / 22 / 472	0.751 [0.679, 0.815]	0.751 [0.679, 0.815]	78.8% vs 18.8%	4.19
DeepSeek V4 Pro, framed, top-3 × HHEM-2.1-Open, top-3	1,221	78 / 151 / 308 / 684	0.025 [-0.033, 0.086]	0.024 [-0.031, 0.081]	34.1% vs 31.6%	1.08
Claude Sonnet 5.5, framed, top-3 × HHEM-2.1-Open, top-3	800	62 / 105 / 209 / 424	0.035 [-0.041, 0.116]	0.033 [-0.038, 0.109]	37.1% vs 33.9%	1.10

Table 13: Error correlation between judges on the items both answered, top-3 condition, held-out motions: counts of items both got wrong / only A / only B / neither, ϕ and κ between the error indicators with bootstrap intervals, and $P(B \text{ wrong} \mid A \text{ wrong})$ against B's own error rate.

set	k	items	mean ϕ	effective votes
Jev + Luna Decisions (top-3)	2	1,614	0.406	1.42
Jev + Luna Decisions + Luna chat + DeepSeek Pro	4	1,221	0.484	1.63
Luna chat + DeepSeek Pro	2	1,221	0.728	1.16
Luna chat + DeepSeek Pro + Claude	3	612	0.738	1.21
Jev + Luna Decisions + Luna chat + DeepSeek Pro + Claude	5	612	0.538	1.59
all six, top-3	6	612	0.362	2.13

Table 14: Effective independent votes of judge sets, $k/(1 + (k-1)\bar{\phi})$, on the items every judge of the set answered.

pending) matters most. The decision model's weakness is concentrated in identifiable types, which is where an LLM second opinion can justify its price.

Kind of judge (H2). Table 13 gives every pair. The mean ϕ is 0.716 between two LLMs, 0.456 between a decision model and an LLM and 0.406 between the two decision models, in the predicted order and with disjoint intervals between the extremes: H2 is supported. Two reasoning models of different families fail on the same items (Luna chat and DeepSeek V4 Pro $\phi = 0.728$: when one is wrong the other is wrong 77.3% of the time against a base rate of 18.8%; DeepSeek and Claude 0.751), far more than the two decision models of different makers do ($\phi = 0.406$, 70.2% against 33.8%). HHEM's errors are uncorrelated with every other judge's (ϕ from -0.066 to 0.035), the signature of a different kind of model, but at its accuracy the independence buys little. Table 14 turns the correlations into effective votes: two LLMs are worth 1.16 independent votes, Jev with Luna Decisions 1.42, the four together 1.63 and all six 2.13. The kind of judge, and not only its accuracy, determines what a second opinion is worth. The LLM pairs here are far more dependent than the 0.21 average error correlation that Hossain et al. [2026] report across ten judges on fixed items, in the direction they note for frontier judges of different providers; the exploratory split below attributes part of the excess to the evidence the judges share. The effective votes agree with Kohli [2026], who get about two votes from nine judges: six judges of three kinds give 2.13 here.

Difficulty (H4). Table 15 repeats the correlations within the two strata. Every correlation falls once items are split by a third judge's success, as it must when part of what judges share is the difficulty of the item. Jev's correlations with the LLMs stay below the Luna chat–DeepSeek pair in both strata (0.223, 0.129 and 0.269 against 0.427 where the third judge is right; 0.184, 0.244 and 0.426 against 0.636 where it is wrong), but not below the Luna chat–Claude pair, which drops to the same level (0.179 and 0.409). H4 is partly supported: the ordering survives stratification for one of the two LLM pairs that can be stratified.

pair	third judge	ϕ, third judge right	ϕ, third judge wrong
Jev, framed, top-3 × Luna Decisions, framed, top-3	DeepSeek V4 Pro, framed, top-3	0.212 [0.137, 0.285] (n 992)	0.370 [0.252, 0.495] (n 229)
Jev, framed, top-3 × Luna chat, framed, top-3	DeepSeek V4 Pro, framed, top-3	0.223 [0.118, 0.327] (n 992)	0.184 [0.046, 0.317] (n 229)
Jev, framed, top-3 × DeepSeek V4 Pro, framed, top-3	Claude Sonnet 5.5, framed, top-3	0.129 [0.005, 0.256] (n 497)	0.244 [0.046, 0.450] (n 115)
Jev, framed, top-3 × Claude Sonnet 5.5, framed, top-3	DeepSeek V4 Pro, framed, top-3	0.269 [0.110, 0.414] (n 494)	0.426 [0.238, 0.586] (n 118)
Luna Decisions, framed, top-3 × Luna chat, framed, top-3	DeepSeek V4 Pro, framed, top-3	0.108 [0.038, 0.182] (n 992)	0.249 [0.106, 0.387] (n 229)
Luna Decisions, framed, top-3 × DeepSeek V4 Pro, framed, top-3	Claude Sonnet 5.5, framed, top-3	0.064 [-0.029, 0.161] (n 497)	0.339 [0.146, 0.519] (n 115)
Luna Decisions, framed, top-3 × Claude Sonnet 5.5, framed, top-3	DeepSeek V4 Pro, framed, top-3	0.062 [-0.025, 0.163] (n 494)	0.353 [0.161, 0.526] (n 118)
Luna chat, framed, top-3 × DeepSeek V4 Pro, framed, top-3	Claude Sonnet 5.5, framed, top-3	0.427 [0.253, 0.591] (n 497)	0.636 [0.437, 0.810] (n 115)
Luna chat, framed, top-3 × Claude Sonnet 5.5, framed, top-3	DeepSeek V4 Pro, framed, top-3	0.179 [0.018, 0.353] (n 494)	0.409 [0.200, 0.596] (n 118)

Table 15: H4: error correlation within difficulty strata, items split by whether a third judge was right.

judge	AUROC, relation question	AUROC, framed criterion	gain
GPT-6 Luna Decisions, 20 passages	0.935 [0.919, 0.949]	0.811 [0.778, 0.842]	-0.124
GPT-6 Luna Decisions, top-3	0.871 [0.850, 0.894]	0.747 [0.714, 0.781]	-0.125
Jev, 20 passages (pilot)	0.777 [0.753, 0.801]	0.980 [0.970, 0.987]	0.203

Table 16: H5: AUROC on numeric held-out items of the arena's relation question against the framed criterion, same input text, for the second decision model; Jev from the pilot for reference.

Framing (H5). Table 16. H5 is not supported, and the sign is reversed. For GPT-6 Luna Decisions the arena's relation question gives AUROC 0.935 against 0.811 for the framed criterion over twenty passages (0.871 against 0.747 over three), a loss of 0.12, while for Jev on the same held-out items the framed criterion is ahead by 0.203. The framing effect of §6.3 is real and large, but it belongs to the question and the model together: the framing that unlocks one decision model handicaps another of a different maker, so a criterion text has to be tuned against the judge that will answer it, with free labels of the kind DebateCheck provides. Liu [2026] report the same sensitivity from the other side: logically equivalent presentations of one grading request change Jev's vulnerability to a one-character edit in the candidate.

Coverage at guaranteed risk. Table 17. Only Jev's probabilities support a rule of this kind: over twenty passages the region fitted on the training motions accepts 44.2% of held-out items at 5.0% error (one-sided 95% bound 6.6%, just above the 5% target), over three passages 25.9% at 3.8% (bound 5.8%). No region of Luna Decisions' or HHEM's probabilities reaches a 5% bound on the training motions, and no agreement rule between two judges gets under 10% error on the items they agree on: judges that agree are often wrong together, which is the correlation of Table 13 seen from the other side. That Luna Decisions' probabilities support no guaranteed-risk region matches the finding of Duan et al. [2026] that decision models tend to identify the most likely outcome while overstating its probability; Jev's probabilities, on this task, are usable as confidence.

Task objectivity (H1). Tables 18, 20 and 19. H1 is not supported. On CHARM-100, DebateCheck's ϕ is the lower one for 4 of 6 pairs, with disjoint intervals for 3, short of the 80% the pre-registration asked for; on HealthBench-100 it is the lower one for one pair of three. The pattern is not noise. The pairs that include a decision model are less correlated on the objective task than on the panel (Jev–Luna Decisions 0.406 against 0.626, Luna Decisions–Luna chat 0.379 against 0.611), while the LLM pair is more correlated on it (Luna chat–Claude 0.668 against 0.577). A task with free labels and fixed evidence does not lower shared error across the board; it lowers it for decision models and raises it for LLMs, which the next paragraph explains. On the panels themselves Jev agrees with the human labels as well as the best LLM on CHARM-100 (accuracy 81.7%, κ 0.595, against 82.9% for Claude and 74.9% for Luna chat) and better than both OpenAI judges on HealthBench-100 (80.5% against 69.3% and 60.0%), which is the finding of Rao and Callison-Burch [2026c] reproduced on their own material.

judge	thresholds (low / high)	coverage	accepted	error among accepted	95% upper bound
Jev, framed, 20 passages	0.16 / 0.88	44.2%	713	5.0%	6.6%
Jev, framed, top-3	0.10 / 0.92	25.9%	418	3.8%	5.8%
Luna Decisions, framed, 20 passages	none at 5%	–	–	–	–
Luna Decisions, framed, top-3	none at 5%	–	–	–	–
HHEM-2.1-Open, top-3	none at 5%	–	–	–	–

Table 17: Coverage at guaranteed risk: acceptance thresholds fitted on the training motions so that the one-sided Clopper–Pearson 95% bound on the error among accepted items is at most 5%, then applied to the held-out motions.

judge	n	coverage	judge MET	label MET	accuracy	κ
Jev (noul framed)	480	97.8%	76.5%	59.0%	81.7% [77.3%, 85.8%]	0.595 [0.495, 0.684]
Jev (choice framing)	480	97.8%	75.8%	59.0%	82.3% [77.9%, 86.3%]	0.610 [0.512, 0.698]
Luna Decisions	491	100.0%	68.0%	59.1%	82.9% [79.0%, 86.6%]	0.634 [0.545, 0.713]
Luna chat	491	100.0%	67.8%	59.1%	74.9% [70.5%, 79.3%]	0.464 [0.364, 0.557]
Claude Sonnet 5.5	490	99.8%	66.9%	59.2%	82.9% [78.9%, 86.5%]	0.634 [0.550, 0.711]

Table 18: CHARM-100 (491 criterion judgements with a human label): agreement of each judge with the human labels under the MET/UNMET reading.

Exploratory: shared evidence, shared errors. This check was not pre-registered. Three passages chosen by BM25 do not always contain the fact: the passage carrying the original token is among them for 77.3% of held-out pairs. Table 21 splits recall by this. Where the fact passage is present, the LLM judges catch most altered sentences (Luna chat 88.1%, DeepSeek 91.5%, Claude 86.1%); where it is absent they fall to 40.4%, 44.4% and 40.2%, and the correlation between two LLMs' errors is 0.732 in that stratum against 0.695 where the fact is present (Table 22). Jev over three passages falls from 80.0% to 39.9%, and over twenty, where the fact passage is always read, stays at 77.6% on the same pairs. Much of what the LLM judges share on DebateCheck is therefore the evidence they were handed: models that read the three passages carefully and find no passage about the fact all answer the same way, and that answer is wrong by construction. Shared error has an evidence component that a judge-only account misses, and that the twenty-passage condition removes.

pair	ϕ on DebateCheck	ϕ on CHARM-100	ϕ on HealthBench-100 test
Jev (noul framed) × Luna Decisions	0.406 [0.357, 0.458]	0.626 [0.521, 0.731]	0.201 [0.059, 0.344]
Jev (noul framed) × Luna chat	0.491 [0.439, 0.543]	0.526 [0.431, 0.616]	0.286 [0.130, 0.443]
Jev (noul framed) × Claude Sonnet 5.5	0.588 [0.512, 0.656]	0.583 [0.483, 0.682]	–
Luna Decisions × Luna chat	0.379 [0.325, 0.432]	0.611 [0.534, 0.688]	0.384 [0.268, 0.515]
Luna Decisions × Claude Sonnet 5.5	0.377 [0.300, 0.452]	0.574 [0.479, 0.662]	–
Luna chat × Claude Sonnet 5.5	0.668 [0.602, 0.731]	0.577 [0.502, 0.645]	–

Table 19: H1: the same pair of judges on DebateCheck (top-3, held-out) and on the subjective panels; ϕ between error indicators with bootstrap intervals.

judge	n	coverage	judge MET	label MET	accuracy	κ
Jev (noul framed)	205	100.0%	52.7%	53.7%	80.5% [75.2%, 86.0%]	0.608 [0.501, 0.716]
Jev (choice framing)	205	100.0%	52.7%	53.7%	78.5% [72.6%, 83.8%]	0.569 [0.451, 0.672]
Luna Decisions	205	100.0%	34.1%	53.7%	60.0% [52.0%, 67.4%]	0.218 [0.076, 0.350]
Luna chat	205	100.0%	36.6%	53.7%	69.3% [62.2%, 76.5%]	0.397 [0.269, 0.528]

Table 20: HealthBench-100 test (205 criterion judgements with a physician majority label): agreement of each judge with the human labels.

judge	recall, fact passage present	recall, absent	false alarm, present	false alarm, absent
Jev, framed, top-3	80.0% [76.9%, 83.2%]	39.9% [32.8%, 47.0%]	12.5%	21.3%
Luna Decisions, framed, top-3	74.2% [70.7%, 77.6%]	38.3% [31.1%, 45.4%]	34.8%	30.1%
Luna chat, framed, top-3	88.1% [85.4%, 90.5%]	40.4% [33.3%, 47.5%]	17.6%	24.6%
DeepSeek V4 Pro, framed, top-3	91.5% [89.0%, 94.2%]	44.4% [36.1%, 52.3%]	19.1%	17.9%
Claude Sonnet 5.5, framed, top-3	86.1% [82.2%, 89.8%]	40.2% [30.9%, 50.5%]	17.2%	15.5%
HHEM-2.1-Open, top-3	83.3% [80.6%, 86.4%]	88.5% [84.2%, 92.9%]	46.3%	68.9%
Jev, framed, 20 passages	83.7% [80.6%, 86.5%]	77.6% [71.6%, 83.6%]	8.7%	15.3%
Luna Decisions, framed, 20 passages	78.4% [75.2%, 81.4%]	74.9% [68.3%, 80.9%]	36.7%	38.8%

Table 21: Exploratory, not pre-registered: held-out items split by whether the passage carrying the original token is among the three passages selected for the altered sentence (present for 77.3% of pairs).

pair	ϕ, fact passage present	ϕ, absent
Jev, framed, top-3 × Luna Decisions, framed, top-3	0.343 [0.282, 0.402] (n 1,248)	0.498 [0.403, 0.595] (n 366)
Jev, framed, top-3 × Luna chat, framed, top-3	0.417 [0.347, 0.484] (n 1,248)	0.522 [0.427, 0.616] (n 366)
Jev, framed, top-3 × DeepSeek V4 Pro, framed, top-3	0.439 [0.342, 0.517] (n 954)	0.601 [0.489, 0.701] (n 267)
Jev, framed, top-3 × Claude Sonnet 5.5, framed, top-3	0.518 [0.416, 0.616] (n 606)	0.643 [0.510, 0.756] (n 194)
Jev, framed, top-3 × HHEM-2.1-Open, top-3	0.099 [0.039, 0.164] (n 1,248)	-0.236 [-0.346, -0.131] (n 366)
Luna Decisions, framed, top-3 × Luna chat, framed, top-3	0.331 [0.269, 0.392] (n 1,248)	0.425 [0.318, 0.529] (n 366)
Luna Decisions, framed, top-3 × DeepSeek V4 Pro, framed, top-3	0.300 [0.226, 0.370] (n 954)	0.496 [0.382, 0.608] (n 267)
Luna Decisions, framed, top-3 × Claude Sonnet 5.5, framed, top-3	0.304 [0.217, 0.401] (n 606)	0.464 [0.334, 0.594] (n 194)
Luna Decisions, framed, top-3 × HHEM-2.1-Open, top-3	-0.000 [-0.055, 0.056] (n 1,248)	-0.307 [-0.413, -0.210] (n 366)
Luna chat, framed, top-3 × DeepSeek V4 Pro, framed, top-3	0.695 [0.623, 0.767] (n 954)	0.732 [0.637, 0.817] (n 267)
Luna chat, framed, top-3 × Claude Sonnet 5.5, framed, top-3	0.606 [0.512, 0.692] (n 606)	0.708 [0.597, 0.811] (n 194)
Luna chat, framed, top-3 × HHEM-2.1-Open, top-3	0.098 [0.039, 0.155] (n 1,248)	-0.235 [-0.331, -0.127] (n 366)
DeepSeek V4 Pro, framed, top-3 × Claude Sonnet 5.5, framed, top-3	0.632 [0.513, 0.739] (n 472)	0.909 [0.832, 0.971] (n 140)
DeepSeek V4 Pro, framed, top-3 × HHEM-2.1-Open, top-3	0.117 [0.045, 0.188] (n 954)	-0.268 [-0.389, -0.145] (n 267)
Claude Sonnet 5.5, framed, top-3 × HHEM-2.1-Open, top-3	0.188 [0.103, 0.277] (n 606)	-0.362 [-0.492, -0.216] (n 194)

Table 22: Exploratory: error correlation within the two evidence strata of Table 21.

7.4 The arena with the certified question

Study 2 certifies one question for the arena's retraction: the framed criterion over every passage the arena read, with Jev deciding at 0.5 or, for a strike that has to be right, at the confident threshold of the guaranteed-risk rule ($p \geq 0.88$, about 5% error on held-out DebateCheck). I judged every claim of the 180 pilot debates again under it: 3,357 claims, $0.38 of judge time, no new debate, because every sentence is stored with the passages it was judged against. The pilot's own question had struck 20 claims. The framed criterion marks 755 at 0.5 (22.5%) and 90 at the confident rule (2.7%).

model	claims	struck (pilot)	contradicted at 0.5	rate	confident ($p \geq 0.88$)	rate	95% interval	pilot win rate
Claude Sonnet 5.5	590	0	111	18.8%	7	1.2%	[0.6%, 2.4%]	61.7%
Claude Opus 5.5	728	1	144	19.8%	15	2.1%	[1.3%, 3.4%]	71.7%
GPT-6.1 Sol	455	0	63	13.8%	10	2.2%	[1.2%, 4.0%]	63.3%
DeepSeek V4 Flash	667	5	209	31.3%	24	3.6%	[2.4%, 5.3%]	43.3%
DeepSeek V4 Pro	580	13	174	30.0%	21	3.6%	[2.4%, 5.5%]	34.2%
GPT-6 Luna	337	1	54	16.0%	13	3.9%	[2.3%, 6.5%]	25.8%

Table 23: Leaderboard v2: the pilot's 3,357 claims judged again by Jev under the framed criterion over every passage the arena read. "Struck (pilot)" is what the arena's original question caught. Models are ordered by the confident rate; Wilson 95% intervals. The win rate is the pilot's match outcome, unchanged.

motion	claims	contradicted at 0.5	rate	confident	rate
congestion-pricing	514	147	28.6%	4	0.8%
crewed-spaceflight	540	64	11.9%	3	0.6%
four-day-week	589	81	13.8%	3	0.5%
nuclear-power	608	161	26.5%	26	4.3%
rent-control	545	111	20.4%	16	2.9%
sugar-tax	561	191	34.0%	38	6.8%

Table 24: Leaderboard v2 by motion: contradiction rates vary more across motions than across models.

Table 23 is the leaderboard the arena promised. Under the confident rule Claude Sonnet 5.5 is struck least (1.2% of its claims, interval [0.6%, 2.4%]), and the three models at the bottom, the two DeepSeek models and GPT-6 Luna (3.6%, 3.6% and 3.9%), cannot be told apart: Luna's 13 strikes on 337 claims give it the highest rate by one or two strikes, with the widest interval ([2.3%, 6.5%]). The ends of the ranking are separated, the neighbours are not. At the 0.5 rule the two DeepSeek models stand apart from the other four (30.0% and 31.3% of their claims against 13.8% to 19.8%), with disjoint intervals, and the four overlap among themselves. The spread the leaderboard was built to show exists, and it broadly follows the pilot's match outcomes at the top (the two Claude models and GPT-6.1 Sol) and the bottom (GPT-6 Luna). Two cautions belong with it. The rate varies more across motions than across models (Table 24): motions whose packs are dense with figures (sugar tax, nuclear power) produce more strikes for everyone, so the comparison holds within a schedule in which every model argued every motion on both sides, as here, and not across schedules. And a strike means that the library gives a different figure for the same thing, which the pilot's hand review showed can be an edition conflict (Lazard's 2023 against 2024 cost ranges, a study quoted from a later revision) rather than a falsehood; the 90 confident strikes are listed with the data for review, and the arena still needs a third verdict, "not in this edition", before a strike can be read as an error of the debater rather than a disagreement with the library.

7.5 What the study says

Of five pre-registered predictions, two held (H2, H3), one held in part (H4) and two failed (H1, H5); the failed predictions carry the most information. Error overlap between judges is ordered

by the kind of judge: two reasoning models of different families are worth 1.16 votes, a decision model and an LLM more, two decision models of different makers most, and an NLI classifier is independent of all of them but too weak to use. The overlap is not lower on a task with free labels in general; it is lower for pairs that include a decision model and higher for pairs of LLMs, and the exploratory split shows why: LLMs that read the same three passages fail together when the fact is not in them, so part of what looks like shared judgement is shared evidence. The framing that moved Jev from catching one altered number in twenty to nine in ten moves a second decision model the other way, so the lesson of §6.3 is not that the framed criterion should be adopted everywhere, but that the question has to be tuned to the judge that answers it, with labels one can afford. Jev's weakness has a shape, units and attributions, where an LLM second opinion is worth having and where the human check of the construction labels is due. And Jev over twenty passages remains the best single judge on this task and the only one whose probability supports a guaranteed-risk rule, while agreeing with human labels on the subjective panels as well as the LLMs do, the result the Autorubric authors reported, reproduced here on their material with their library. With that question the arena does what it was built for: the same 180 debates, judged again for $0.38, give 90 confident strikes instead of 20 and a leaderboard whose ends are separated.

8 Discussion

The criterion text determines the judge. Holding everything else fixed, the way the question is posed moved Jev from catching one altered number in twenty to catching nine in ten. The components are separable: the *wording* of the contradiction option moves both ordering and location; the *evidence selection* moves location more than ordering; the *threshold* is an offset that free labels can fit; and the *framing* (a task sentence, a criterion stated as a proposition about the submission, verdict definitions as the outcomes) moves everything at once. Rao and Callison-Burch [2026c] could only observe that judges share conventions the rubric does not state; here the convention is explicit ("a different number for the same thing counts"), the labels enforce it, and the gap closes.

What this means for the arena. The retraction rate the leaderboard was built on is not a property of the debaters; it is a property of a question. I reran the judge, not the debates: for $0.38 the same 180 debates give 90 confident strikes instead of 20 and the leaderboard of §7.4, because every sentence is stored with the passages it was judged against. The same change makes strict mode, which rewinds a turn when a sentence is struck, worth testing again. What the framed criterion does not fix is the convention problem I found on the sharp wording's false alarms in the pilot: a model that correctly quotes Lazard's 2024 cost ranges when the pack holds the 2026 edition is giving "a different number for the same thing". A fact-checking criterion needs a third verdict, not covered by this edition, and evidence that records the edition and year of each figure.

What this means for decision-model evaluation. Perturbation gives labelled items at no cost, in the register of the deployment, in any quantity, and with a known direction of error. It cannot replace human labels for the question whether an original sentence is true; it can replace them for the question whether a judge reacts to a change that must change the label. I suggest it as a standard check before any decision-model deployment: alter the thing the criterion is about, and see whether the probability moves.

9 Limitations

The positive labels are by construction; the negative labels are a judge's supported verdicts, reviewed by a language model on a sample, not by people. Only numbers were altered; a wrong name, a reversed causal claim or a fabricated study would need other perturbations, and the

two-step and direct forms may rank differently on them. One Jev version was used; its answers drift by about 0.03 between calls, and the API is in early access. The LLM judges ran at low reasoning effort with one prompt. The sample reviews and the reviser in the optimisation loop are outputs of language models, that is, of judges of the kind under study; the released sheets are meant to be checked by a person. Six English-language policy motions and six debaters are a small world, and the held-out split, two motions, is small. Costs are list prices, not invoices. Study 2 adds its own limits: Claude Sonnet 5.5 judged a 400-pair sample, not every pair; DeepSeek V4 Pro has no panel results; the human validation of the new perturbation types was prepared but not yet collected, so the unit type, where every judge is near chance, may contain edits that are not contradictions (percent against percentage points, a figure per year against per month); and the panels hold 100 items each, so their intervals are wide.

10 Conclusion

A live arena in which a decision model retracts sentences during streaming is cheap and fast enough to run; its first tournament showed that the question it asked could not see a wrong number. The same debates, re-used as a source of contrastive pairs, let me hold the model and the evidence fixed and vary only the question, which turned out to be the hidden variable: wording, evidence selection, threshold and framing each move the judge, and the framing recommended by the authors of the tools I used moves it the most. The pairs, the judgments of three judges on them, the optimisation loop and the reports are released so that the experiment the earlier paper could not run can be run again on other criteria.

The second study, pre-registered and reported with its two failed predictions, says what a second opinion is worth: two LLMs of different families share most of their errors on this task, partly because they share the evidence, while a decision model of a different maker is closer to an independent vote; the framing effect is a property of the question and the model together; and the decision model's blind spots have a shape (units, attributions) that tells a cascade where to spend. The pairs, every judge's answer on them, the panel gradings, the pre-registration with its deviations and the two blind sheets for the human check are released with the code. The next steps are set by the limits: the human labels; a sample of pairs built from sentences the judge had not approved, so that the judge does not select its own test; the open-weight decision models of Duan et al. [2026] as a third maker in the panel, which would test whether two decision models are close to independent votes or whether this pair was; and the criterion-optimisation loop run against the second decision model, since the question that suits one does not suit the other.

Reproducibility and cost. Building the pairs is free. Judging all 1,164 pairs under seven wordings with Jev cost $1.10 (26 million input tokens); the optimisation loop $0.40; GPT-6 Luna and DeepSeek V4 Flash on all pairs about $0.30 and $2.50 (the latter almost entirely reasoning tokens); the Autorubric validation under $0.70. The first study, pilot included, stayed under $12 of API credit, about a third of it on the judge. Study 2 cost, at list prices from the tokens the output files carry, about $3.20 on OpenAI (two Decisions runs on all pairs and the relation-question run, Luna chat on all pairs, the panels), $0.52 on TypeSafe (Jev on 1,657 new pairs under two conditions and on both panels), $1.67 on DeepSeek (whose prepaid balance of $6.43 nevertheless ran out before the last 395 pairs, so its actual charges exceed the list estimate) and $4.61 as charged for Claude Sonnet 5.5 ($2.91 for the 400 pairs and $1.70 for the CHARM-100 grading through Autorubric, whose prompt caching brought a $5.28 list-price run down to that figure); HHEM ran on a laptop CPU. The two studies together stayed under $30 of API credit, about a seventh of it on the decision model that did most of the judging.

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A Wordings tried by the optimisation loop

B Agreement between the two LLM judges

deepseek-v4-flash x gpt-6-luna: both wrong on 14.3% of 2080 items; $P(B \mid A) = 86.3\%$ against a base rate of 16.9%.

C The questions the arena asks

```
is_claim:  Does ‘sentence‘ assert a checkable fact about the world (numbers, events, causes),
           rather than an opinion, value judgment, question, or rhetorical framing?
supported: Is ‘sentence‘ supported by the ‘evidence‘?
           true:  a passage states it or directly implies it
           false: no passage addresses it, or a passage contradicts it
relation:  How does the ‘evidence‘ relate to ‘sentence‘?
           supports:     a passage states the claim or directly implies it is true
           contradicts:  a passage states or directly implies the claim is false
           says_nothing: no passage addresses what the sentence asserts
which:     Which evidence passage best supports ‘sentence‘?  (one option per passage, and none)
on_motion: Is ‘sentence‘ relevant to arguing for or against the ‘motion‘?
```

Round	AUROC	Det.	FA	Wording of the `contradicts` option
0	0.899	37.3%	0.8%	a passage gives a different number, date, name, place or outcome for the same thing 'sentence' describes, or states the opposite
1	0.905	44.3%	1.0%	The 'evidence' gives a different number, date, name, place, threshold, or outcome for a fact asserted in 'sentence', or explicitly states its opposite, even if the rest of the sentence agrees with the evidence.
1	0.876	32.5%	0.8%	A passage in 'evidence' describes the same entity, event, measure, and relevant period as 'sentence' but assigns an incompatible value or detail to at least one asserted fact.
1	0.889	15.8%	0.3%	The 'evidence' conflicts with an asserted fact in 'sentence', not merely its tone or interpretation; figures for another measure, unit, period, source edition, or entity, and missing information, do not count as conflicts.
1	0.894	41.0%	0.8%	Compare each factual detail in 'sentence' with passages in 'evidence' about that same fact, and select this option if any passage supplies an incompatible detail, even when other details match; do not treat an unmentioned detail as contradicted.
2	0.903	38.5%	0.8%	A passage in 'evidence' gives a number, date, name, place, threshold, or outcome incompatible with the corresponding detail in 'sentence', or explicitly states the opposite, even if every other detail agrees.
2	0.886	39.3%	0.8%	The 'evidence' describes the same fact as 'sentence' but assigns it an incompatible value or detail; agreement on the general claim does not cancel a mismatch in one asserted detail.
2	0.893	19.8%	0.3%	The 'evidence' conflicts with a fact asserted in 'sentence'; missing details, equivalent units, and figures for a different entity, measure, period, or source edition do not count as conflicts.
2	0.905	47.0%	0.8%	Compare the factual details in 'sentence' with the corresponding facts in 'evidence' and choose this option if even one detail is incompatible, but not merely because a detail is absent or a passage discusses something different.
3	0.905	49.0%	2.0%	A passage in 'evidence' states a different number, date, name, place, threshold, or outcome for a fact asserted in 'sentence', even when it confirms the main claim or every other detail.
3	0.895	36.8%	0.5%	The 'evidence' and 'sentence' describe the same fact but give incompatible details, such as different start dates for the same scheme or different percentages for the same result; one such mismatch is enough.
3	0.885	26.3%	0.5%	The 'evidence' gives an incompatible detail for the same fact in 'sentence', not merely a missing detail, an equivalent value in different units, or a figure for another measure, period, entity, or source edition.
3	0.901	35.0%	1.0%	Check each asserted detail in 'sentence' against passages in 'evidence' about that fact, and choose this option when any corresponding detail cannot be true alongside the evidence, regardless of other agreements; an unmentioned detail alone does not qualify.

Table 25: Every *contradicts* description the loop evaluated on the 400-pair training sample, with its AUROC and its detection and false-alarm rates at 0.7. Round 0 is the seed.